

We have 250 events in total in sysmon log

I would start by filtering what EVENTID = 1 → means new process creation

```
source="windows-sysmon.log" host="si-i-0ca3967fa13.prd-p-tjo7h.splunkcloud.com" sourcetype="sysmon logs"
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| search EventID=1
```

Now we have down to 25 events

The screenshot shows the Splunk search results for the query. The search bar contains the query: `source='windows-sysmon.log' host='si-i-0ca3967fa13.prd-p-tjo7h.splunkcloud.com' sourcetype='sysmon logs' | rex field=_raw '<EventID>(?!<EventID>\d+)</EventID>' | search EventID=1`. The results table shows 25 events. The first event is at 7/8/26 11:52:52.000 PM. The event details include fields like EventID, Time, Image, and CommandLine.

Time	Event
7/8/26 11:52:52.000 PM	<Event xmlns='http://schemas.microsoft.com/win/2004/08/events/event'><System><Provider Name='Microsoft-Windows-Sysmon' Guid='{5770385F-C22A-43E0-BF4C-06F5698FFB09}' /><EventID>1</EventID><Version>5</Version><Level>4</Level><Task>1</Task><Opcode>0</Opcode><Keywords>0x80000000</Keywords><TimeCreated SystemTime='2021-01-22T20:44:00.340564600Z' /><EventRecordID>31194</EventRecordID><Correlation><Execution ProcessID='2448' ThreadID='3264' /><Channel>Microsoft-Windows-Sysmon/Operational</Channel><Computer>win-dc-770.attackrange.local</Computer><Security UserID='S-1-5-18' /></System><EventData><Data Name='RuleName'>-</Data><Data Name='UtcTime'>2021-01-22 20:44:00.338</Data><Data Name='ProcessGuid'>{732C744F-3910-600B-3E0C-00000000A301}</Data><Data Name='ProcessId'>4420</Data><Data Name='Image'>C:\Windows\System32\cmd.exe</Data><Data Name='FileVersion'>10.0.14393.0 (rs1_release.160715-1616)</Data><Data Name='Description'>Windows Command Processor</Data><Data Name='Product'>Microsoft® Windows® Operating System</Data><Data Name='Company'>Microsoft Corporation</Data><Data Name='OriginalFileName'>Cmd.Exe</Data><Data Name='CommandLine'>"C:\Windows\system32\cmd.exe" /c "wbadmin delete systemstatebackup -keepVersions:0" </Data><Data Name='CurrentDirectory'>C:\Users\ADMINI~1\AppData\Local\Temp\</Data><Data Name='User'>ATTACKRANGE\Administrator</Data><Data Name='LogonGuid'>{732C744F-17E1-600B-DAA7-130000000000}</Data><Data Name='LogonId'>0x13a7da</Data><Data Name='TerminalSessionId'>2</Data><Data Name='IntegrityLevel'>High</Data><Data Name='Hashes'>MD5=F4F684066175B77E0C3A000549D2922C, SHA256=935C1861DF1F4018D698E8B65ABFA02D7E9037D8F68CA3C206586CA165D44AD2, IMPHASH=3062ED732D4B25D1C64F084D97D37A</Data><Data Name='ParentProcessGuid'>{732C744F-1C72-600B-C805-00000000A301}</Data><Data Name='ParentProcessId'>3346</Data><Data Name='ParentImage'>C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe</Data><Data Name='ParentCommandLine'>"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" </Data></EventData></Event>

To quickly understand what was happening, I extracted the Image and CommandLine fields. These fields show which programs were executed and what commands were run.

```
source="windows-sysmon.log" sourcetype="sysmon logs"
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>[^\<]+)</Data>"
| rex field=_raw "<Data Name='Image'>(?!<Image>.*?)</Data>"
| search EventID=1
| table CommandLine Image
```

This allowed me to quickly review the 25 process creation events and look for suspicious activity side by side in table

✓ 25 events (before 7/9/26 12:12:30.000 AM) No Event Sampling		Job		→	↓	Policy-Based Pool	Smart Mode
Events Patterns Statistics (25) Visualization							
Show: 20 Per Page Format Preview: On		< Prev 1 2 Next >					
CommandLine	Image						
"C:\Windows\system32\cmd.exe" /c "wbadmin delete catalog -quiet"	C:\Windows\System32\cmd.exe						
wbadmin delete catalog -quiet	C:\Windows\System32\wbadmin.exe						
"C:\Windows\system32\wbengine.exe"	C:\Windows\System32\wbengine.exe						
C:\Windows\System32\vdslldr.exe -Embedding	C:\Windows\System32\vdslldr.exe						
"C:\Windows\system32\cmd.exe" /c "bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures & bcdedit.exe /set {default} recoveryenabled no"	C:\Windows\System32\cmd.exe						
bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures	C:\Windows\System32\bcdedit.exe						
bcdedit.exe /set {default} recoveryenabled no	C:\Windows\System32\bcdedit.exe						
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" & (Get-WmiObject Win32_Shadowcopy ForEach-Object {\$_.Delete()});	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe						
"C:\Windows\system32\cmd.exe" /c "del /s /f /q c:*.VHD c:*.bac c:*.bak c:*.wbc c:*.bkf c:\Backup*. * c:\back*. * c:*.set c:*.win c:*.dsk"	C:\Windows\System32\cmd.exe						
"C:\Windows\system32\HOSTNAME.EXE"	C:\Windows\System32\HOSTNAME.EXE						
"C:\Windows\system32\whoami.exe"	C:\Windows\System32\whoami.exe						
"C:\Windows\system32\cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"	C:\Windows\System32\cmd.exe						
vssadmin.exe delete shadows /all /quiet	C:\Windows\System32\vssadmin.exe						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe						
C:\Windows\system32\vssvc.exe	C:\Windows\System32\VSSVC.exe						
"C:\Windows\system32\cmd.exe" /c "wmic.exe shadowcopy delete"	C:\Windows\System32\cmd.exe						
wmic.exe shadowcopy delete	C:\Windows\System32\wbem\WMIC.exe						
"C:\Windows\system32\cmd.exe" /c "wbadmin delete systemstatebackup -keepVersions:0"	C:\Windows\System32\cmd.exe						
wbadmin delete systemstatebackup -keepVersions:0	C:\Windows\System32\wbadmin.exe						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	C:\Program Files\SplunkUniversalForwarder\bin\splunk-						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-admon.exe						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-regmon.exe						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-netmon.exe						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe						
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-winprintmon.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-winprintmon.exe						

We already see suspicious command-line executions that follow the pattern of MITRE ATT&CK T1490 below:

- vssadmin delete shadows /all /quiet** → Deletes all Volume Shadow Copies, preventing file recovery.
- wmic shadowcopy delete** → Deletes Volume Shadow Copies using Windows Management Instrumentation (WMI).
- wbadmin delete catalog -quiet** → Deletes the Windows Backup Catalog.
- wbadmin delete systemstatebackup -keepVersions:0** → Deletes system state backups.
- bcdedit /set {default} recoveryenabled no** → Disables the Windows Recovery Environment.
- bcdedit /set {default} bootstatuspolicy ignoreallfailures** → Prevents Windows from entering recovery mode after boot failures.
- `del /s /f /q C:\\*.bak`** → Deletes backup files from the system.

These commands would be IOCs because they are commonly used by attackers to prevent recovery after an attack.

The next step is identifying which user performed the activity.

Events	Patterns	Statistics (29)	Visualization
Show: 20 Per Page	Format	Preview: On	< Prev 1 2 Next >
CommandLine	Image	User	
"C:\Windows\system32\cmd.exe" /c "wbadmin delete catalog -quiet"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator	
wbadmin delete catalog -quiet	C:\Windows\System32\wbadmin.exe	ATTACKRANGE\Administrator	
"C:\Windows\system32\wbengine.exe"	C:\Windows\System32\wbengine.exe	NT AUTHORITY\SYSTEM	
C:\Windows\System32\vdslldr.exe -Embedding	C:\Windows\System32\vdslldr.exe	NT AUTHORITY\SYSTEM	
"C:\Windows\system32\cmd.exe" /c "bcdedit.exe /set (default) bootstatuspolicy ignoreallfailures & bcdedit.exe /set (default) recoveryenabled no"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator	
bcdedit.exe /set (default) bootstatuspolicy ignoreallfailures	C:\Windows\System32\bcdedit.exe	ATTACKRANGE\Administrator	
bcdedit.exe /set (default) recoveryenabled no	C:\Windows\System32\bcdedit.exe	ATTACKRANGE\Administrator	
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" & {Get-WmiObject Win32_Shadowcopy ForEach-Object {\$_.Delete();}}	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	ATTACKRANGE\Administrator	
"C:\Windows\system32\cmd.exe" /c "del /s /f /q c:*.VHD c:*.bac c:*.bak c:*.wcat c:*.bkf c:\Backup*. * c:\backup*. * c:*.set c:*.win c:*.dsk"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator	
"C:\Windows\system32\HOSTNAME.EXE"	C:\Windows\System32\HOSTNAME.EXE	ATTACKRANGE\Administrator	
"C:\Windows\system32\whoami.exe"	C:\Windows\System32\whoami.exe	ATTACKRANGE\Administrator	
"C:\Windows\system32\cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator	
vssadmin.exe delete shadows /all /quiet	C:\Windows\System32\vssadmin.exe	ATTACKRANGE\Administrator	
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe"	C:\Program Files\SplunkUniversalForwarder\bin\splunk-MonitorNoHandle.exe	NT AUTHORITY\SYSTEM	
C:\Windows\system32\vssvc.exe	C:\Windows\System32\VSSVC.exe	NT AUTHORITY\SYSTEM	
"C:\Windows\system32\cmd.exe" /c "wmic.exe shadowcopy delete"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator	
wmic.exe shadowcopy delete	C:\Windows\System32\wbem\WMIC.exe	ATTACKRANGE\Administrator	
"C:\Windows\system32\cmd.exe" /c "wbadmin delete systemstatebackup -keepVersions:0"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator	
wbadmin delete systemstatebackup -keepVersions:0	C:\Windows\System32\wbadmin.exe	ATTACKRANGE\Administrator	
"C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe" --ps2	C:\Program Files\SplunkUniversalForwarder\bin\splunk-powershell.exe	NT AUTHORITY\SYSTEM	

After reviewing the events, It looks like we have the ATTACKRANGE\Administrator

```
source="windows-sysmon.log" sourcetype="sysmon logs"
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>[^\<]+)</Data>"
| rex field=_raw "<Data Name='Image'>(?!<Image>.*?)</Data>"
| rex field=_raw "<Data Name='User'>(?!<User>.*?)</Data>"
| search EventID = 1 User="ATTACKRANGE\Administrator"
| table CommandLine Image User
```

New Search

source="windows-sysmon.log" sourcetype="sysmon logs"

| rex field=_raw "<EventID>(?(EventID\d+)</EventID>"

| rex field=_raw "<Data Name='CommandLine'>(?(CommandLine>[^<]+)</Data>"

| rex field=_raw "<Data Name='Image'>(?(Image>.*?)</Data>"

| rex field=_raw "<Data Name='User'>(?(User>.*?)</Data>"

| search EventID=1 User="ATTACKRANGE\Administrator"

| table CommandLine Image User

Save As

Create Table View

Close

Time range: All time

15 events (before 7/9/26 12:33:53.000 AM)

No Event Sampling

Job

||

Policy-Based Pool

Smart Mode

Events

Patterns

Statistics (15)

Visualization

Show: 20 Per Page

Format

Preview: On

CommandLine	Image	User
"C:\Windows\system32\cmd.exe" /c "wbadmin delete catalog -quiet"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
wbadmin delete catalog -quiet	C:\Windows\System32\wbadmin.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures & bcdedit.exe /set {default} recoveryenabled no"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures	C:\Windows\System32\bcdedit.exe	ATTACKRANGE\Administrator
bcdedit.exe /set {default} recoveryenabled no	C:\Windows\System32\bcdedit.exe	ATTACKRANGE\Administrator
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" & (Get-WmiObject Win32_Shadowcopy ForEach-Object {\$_Delete()});	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "del /s /f /q c:*.VHD c:*.bac c:*.bak c:*.wbcat c:*.bkf c:\Backup*. c:\Backup*. c:*.set c:*.win c:*.dsk"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\HOSTNAME.EXE"	C:\Windows\System32\HOSTNAME.EXE	ATTACKRANGE\Administrator
"C:\Windows\system32\whoami.exe"	C:\Windows\System32\whoami.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
vssadmin.exe delete shadows /all /quiet	C:\Windows\System32\vssadmin.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "wmic.exe shadowcopy delete"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
wmic.exe shadowcopy delete	C:\Windows\System32\wbem\WMIC.exe	ATTACKRANGE\Administrator

We have down to 15 events

I then reviewed all activity related to this account to make sure these events were connected to process creation activity.(EVENT ID ==1) do not miss anything related the attack

```
source="windows-sysmon.log" sourcetype="sysmon logs"
| rex field=_raw "<EventID>(?(EventID\d+)</EventID>"
| rex field=_raw "<Data Name='CommandLine'>(?(CommandLine>[^<]+)</Data>"
| rex field=_raw "<Data Name='Image'>(?(Image>.*?)</Data>"
| rex field=_raw "<Data Name='User'>(?(User>.*?)</Data>"
| search User="ATTACKRANGE\Administrator"
| table CommandLine Image User
```

splunk> Cloud

New Search

```
source="windows-sysmon.log" sourcetype="sysmon logs"
| rex field=_raw "<EventID>{<EventID>\d+}</EventID>"
| rex field=_raw "<Data Name='CommandLine'>{<CommandLine>.*}</Data>"
| rex field=_raw "<Data Name='Image'>{<Image>.*}</Data>"
| rex field=_raw "<Data Name='User'>{<User>.*}</Data>"
| search User="ATTACKRANGE\Administrator"
| table CommandLine Image User
```

15 events (before 7/9/26 12:34:58.000 AM) No Event Sampling

Events Patterns Statistics (15) Visualization

Show: 20 Per Page Format Preview: On

CommandLine	Image	User
"C:\Windows\system32\cmd.exe" /c "wbadmin delete catalog -quiet"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
wbadmin delete catalog -quiet	C:\Windows\System32\wbadmin.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures & bcdedit.exe /set {default} recoveryenabled no"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures	C:\Windows\System32\bcdedit.exe	ATTACKRANGE\Administrator
bcdedit.exe /set {default} recoveryenabled no	C:\Windows\System32\bcdedit.exe	ATTACKRANGE\Administrator
"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" & (Get-WmiObject Win32_Shadowcopy ForEach-Object {\$_Delete();})	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "del /s /f /q c:*.VHD c:*.bac c:*.bak c:*.wbcat c:*.bkf c:\Backup*. c:\backup*. c:*.set c:*.win c:*.dsk"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\HOSTNAME.EXE"	C:\Windows\System32\HOSTNAME.EXE	ATTACKRANGE\Administrator
"C:\Windows\system32\whoami.exe"	C:\Windows\System32\whoami.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator
vssadmin.exe delete shadows /all /quiet	C:\Windows\System32\vssadmin.exe	ATTACKRANGE\Administrator
"C:\Windows\system32\cmd.exe" /c "wmic.exe shadowcopy delete"	C:\Windows\System32\cmd.exe	ATTACKRANGE\Administrator

This confirms that all action related tot he account ATTACKRANGE\Administrator are only 15 events

As a Tier 1 SOC Analyst, my next step would be to validate the activity, collect all relevant evidence, and escalate the incident to the Tier 2 SOC team for further investigation. If I have containment privileges and the attack is still active, I would isolate the affected endpoint and stop the malicious processes to prevent further impact.

If this investigation contained thousands of Event ID 1 events (not just 25) ,then this case I would search for patterns related to MITRE ATT&CK T1490 beow

Common commands:

vssadmin
wmic
wbadmin
bcdedit
diskshadow
Powershell

Common keywords:

delete
shadow
shadowcopy
shadows
backup
catalog
recovery
recoveryenabled
Bootstatuspolicy

New Search

source="windows-sysmon.log" sourcetype="sysmon logs"

```
| rex field=_raw "<EventID>{<EventID>d+}</EventID>"
| rex field=_raw "<Data Name='CommandLine'>{<CommandLine>[<]*}</Data>"
| rex field=_raw "<Data Name='Image'>{<Image>.}*</Data>"
| rex field=_raw "<Data Name='User'>{<User>.}*</Data>"
| search EventID=1
| search CommandLine="vssadmin*"
    OR CommandLine="wmic*"
    OR CommandLine="wbadmin*"
    OR CommandLine="bcdedit*"
    OR CommandLine="diskshadow*"
    OR CommandLine="powershell*"
| search CommandLine="delete*"
    OR CommandLine="shadows*"
    OR CommandLine="backup*"
    OR CommandLine="catalog*"
    OR CommandLine="recovery*"
    OR CommandLine="recoverycanable*"
    OR CommandLine="bootstatuspolicy*"
| table _time User Image CommandLine
```

✓ **12 events** (before 7/9/26 12:47:14.000 AM) [No Event Sampling]

[Job] [Pause] [Refresh] [Download] [Policy-Based Pool] [Smart Mode]

Events Patterns **Statistics (12)** Visualization

Show: 20 Per Page Format Preview: On

_time ↕	User ↕	Image ↕	CommandLine ↕
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\cmd.exe	"C:\Windows\system32\cmd.exe" /c "wbadmin delete catalog -quiet"
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\wbadmin.exe	wbadmin delete catalog -quiet
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\cmd.exe	"C:\Windows\system32\cmd.exe" /c "bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures & bcdedit.exe /set {default} recoveryenabled no"
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\bcdedit.exe	bcdedit.exe /set {default} bootstatuspolicy ignoreallfailures
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\bcdedit.exe	bcdedit.exe /set {default} recoveryenabled no
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" & {Get-WmiObject Win32_Shadowcopy ForEach-Object {\$_.Delete()}}
2026-07-08 23:52:52	ATTACKRANGE\Administrator	C:\Windows\System32\cmd.exe	"C:\Windows\system32\cmd.exe" /c "vssadmin.exe delete shadows /all /quiet"

```
source="windows-sysmon.log" sourcetype="sysmon logs"
| rex field=_raw "<EventID>(?!<EventID>\d+)</EventID>"
| rex field=_raw "<Data Name='CommandLine'>(?!<CommandLine>[^<]+)</Data>"
| rex field=_raw "<Data Name='Image'>(?!<Image>.*?)</Data>"
| rex field=_raw "<Data Name='User'>(?!<User>.*?)</Data>"
| search EventID=1
| search CommandLine="*vssadmin*"
  OR CommandLine="*wmic*"
  OR CommandLine="*wbadmin*"
  OR CommandLine="*bcdedit*"
  OR CommandLine="*diskshadow*"
  OR CommandLine="*powershell*"
| search CommandLine="*delete*"
  OR CommandLine="*shadow*"
  OR CommandLine="*backup*"
  OR CommandLine="*catalog*"
  OR CommandLine="*recovery*"
  OR CommandLine="*recoveryenabled*"
  OR CommandLine="*bootstatuspolicy*"
| table time User Image CommandLine
```

Thus right here we the 12 events related to T1490.